

(notes) 6.1 Exponential functions

Exponential form: $f(x) = ab^x$

Exponential functions can either be strictly increasing (growth function) or strictly decreasing (decay function). The variable x is in the exponents and will indicate a number repeated multiplications of b .

Note, exponents are shorthand for repeated multiplication while multiplication is shorthand for repeated addition.

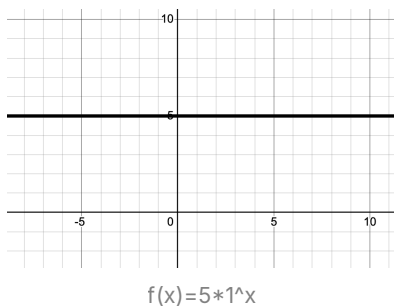
The a is the starting amount.

The b is the proportion that changes every timestep.

Constant function

If $b = 1$, there is no growth and we have a constant function.

ex. $f(x) = 1^x$. It doesn't matter how many repeated multiplications, the overall function value will not change!



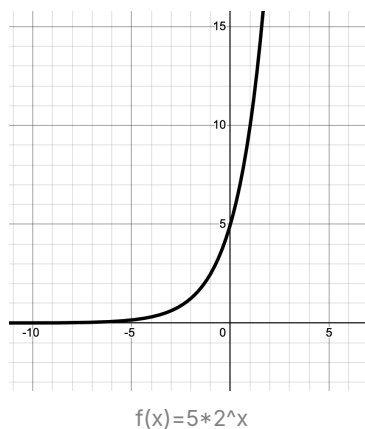
Increasing function

If $b > 1$, there will be growth

ex. $f(x) = 2^x$, every increase in x , we will double!

ex. A disease has a spread rate of 15%:

$$I(t) = (1 + 0.15)^x = 1.15^x$$

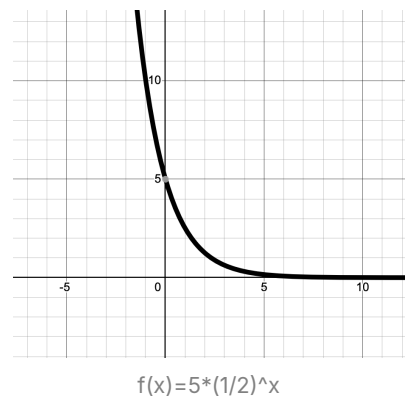


Decreasing function

If $b < 1$, there will be decay

ex. $f(x) = (\frac{1}{2})^x$, every time we increment on x , the value halves

ex. Population decreases by 10% every year: $P(t) = (1 - 0.10)^x = 0.90^x$



The x is the number of timesteps.

Exponential form: $f(x) = ab^x$

Creating functions from word problems

Example. Write the b , proportion, and a , starting amount for the following equations. Answer the questions.

1. Your friend gives you ONE kombucha scoby that doubles every day.
 - function:
 - How many scobies will you have after 1 week?
 - How long until you have 100 scobies? (spoiler: you can answer this question using the inverse of exponential functions which is the logarithmic function. Or you can plot it.)
2. You buy now pay later \$100 with Klarna. You do not pay and so your account was sent to collections where they charge an interest rate of 25% a year.
 - function:
 - How much will you owe after college (after 4 years)
 - How long will it take for you to owe \$1000???
3. Your grandma gave you \$100 for your birthday. You forgot about it and now, 15 years later, how much **buying power** does the \$100 have compared to 15 years ago? Every year, there has been 5% inflation and so your money has depreciated in value 😞
 - function:
 - How much was the \$100 worth after 5 years? After 15 years?
 - How long until the \$100 she gave you can only buy \$5 of stuff?

Creating functions from 2 points.

ex. An exponential function $f(x) = ab^x$ passes through the points $(0, 12)$ and $(3, 96)$. Solve for the starting amount a and the proportion increase/decrease, b . Then, write the full function.

Using the first point, $(0, 12)$, since $x = 0$, this is the initial/starting value. We can immediately write that $a = 12$ and fill that in our equation:

$$\begin{aligned} y &= 12b^x \\ 96 &= 12b^3 && \rightarrow \text{use the point } (3, 96) \text{ to fill in } x = 3, y = 96 \\ \frac{96}{12} &= \frac{12b^3}{12} && \rightarrow \text{divide by 12 to solve for } b \\ \sqrt[3]{8} &= \sqrt[3]{b^3} && \rightarrow \text{cube root} \\ 2 &= b && \implies f(x) = 12 \cdot 2^x \end{aligned}$$

Practice.

1. An exponential function $f(x) = ab^x$ passes through the points $(0, 8)$ and $(2, 128)$. Solve for the starting amount a and the proportion increase/decrease, b . Then, write the full function.
2. An exponential function $f(x) = ab^x$ passes through the points $(0, 6)$ and $(2, 150)$. Solve for the starting amount a and the proportion increase/decrease, b . Then, write the full function.
3. An exponential function $f(x) = ab^x$ passes through the points $(0, 7000)$ and $(3, 3584)$. Solve for the starting amount a and the proportion increase/decrease, b . Then, write the full function.
(answer: $a = 7000, b = 0.8$)
4. An exponential function $f(x) = ab^x$ passes through the points $(0, 5000)$ and $(3, 1080)$. Solve for the starting amount a and the proportion increase/decrease, b . Then, write the full function.
(answer: $a = 5000, b = 0.6$)

Compounding Interest Formulas

Discrete compounding

$$A(t) = P \left(1 + \frac{r}{n} \right)^{nt}$$

$A(t)$: ending amount after t time

t : normally measured in years

P : starting amount, AKA "principal"

r : the **rate per year**, the Annual Percentage Rate (APR)

n : number of compoundings a year....the more compounding=more money!

Continuous compounding

$$A(t) = Pe^{rt}$$

$A(t)$: ending amount after t time

t : normally measured in years

P : starting amount, AKA "principal"

r : the **rate per year**, the Annual Percentage Rate (APR)

n :not in this equation because n is infinity due to the continuous compounding = MOST money!

Ex. You invest \$2000 in a bank account at an interest rate of 3.5% APR. Write the equations and find the ending balance after 5 years if the interest is compounded...

- annually (1 time a year):
- quarterly (4 times a year):
- monthly (____ times a year):
- continuously: